Automated Testing and Deployment with GitHub Actions

Software Engineering Summer School
June 2026

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Automated Testing and Deployment

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Continuous Integration/Continuous Deployment (CI/CD)

What is CI/CD?

- Continuous Integration (CI)
    - Frequent merging of changes into main branch with active quality verification

- Continuous Deployment (CD)
    - Automatic release of new software versions

What is automation for?

- Building documentation
  - Run Sphinx/Doxygen/etc to generate documentation in pdf or html form
  - Check if examples in the documentation work
- Build static websites
  - Build and publish a website to GitHub Pages/ReadTheDocs/etc
- Generating pull requests for updates
  - Update dependencies
  - Fix typos and coding style
- Whatever your project needs!

What are the benefits?

- Consistent, controlled environment between runs
- Runs every PR/commit/tag/whatever you choose
- Can’t be skipped for forgotten, no contributor setup
- Can run lots of OS’s, Python versions, compilers, etc.

Some major CI services

- **Travis CI**: Very popular for years, but not anymore.

- **Jenkins**: A self-host only OSS solution.

- **Circle CI**: The first more “modern” design.

- **GitLab CI**: For years, this was one of the best services. Still very good.

- **Azure Pipelines**: Very modular design is easy to upgrade and maintain.

- **GitHub Actions (GHA)**: Extremely simple and popular. Actions are easy to write and share.

Today we will be focusing on GitHub Actions

We first need to briefly discuss exit codes and YAML

Exit Codes

- Every time you run a command in a shell there is an exit code that indicates if it ran successfully, or if there was an error.
- An exit code of `0` indicates that the command ran successfully, other numbers indicate an error.
- Sometimes different numbers correspond to different errors.

Exit Codes

```
> mkdir test
> echo $?
0
```

```
> mkdir test
mkdir: test: File exists
> echo $?
1
```

```
> mkdir -z test
mkdir: illegal option - z
usage: mkdir [-pv] [-m mode] directory_name ...
> echo $?
64
```

Exit Codes

- CI workflows will typically stop once they encounter a non-zero exit code.
- Sometimes you may need to run a command that might fail, but you want the workflow to proceed.
- Some scripts and binaries don’t respect this standard and return non-zero exit codes even when successful.

Exit Codes

- Error codes can be ignored using logical Or (||)

```
> mkdir -z test || echo "ignore"
ignore
```

YAML

YAML (YAML Ain’t Markup Language, originally Yet Another Markup Language) is a human-readable data serialization language.

- Easy to read and use
- Very commonly used in CI configuration files
- File extension is .yml or .yaml

YAML: Defining Scalar Values

```yaml
number-value: 42
boolean-value: true # can also be on or yes
string-value: "Hello world"
another-string: String without quotes
```

YAML: Defining lists

```yaml
colors:
  - red
  - green
  - blue

more_colors: [black, white]
```

YAML: Defining dictionaries

```yaml
person:
  name: John Smith
  age: 33
  occupation: accountant

same_person: {name: John Smith, age: 33, occupation: accountant}
```

YAML: Defining multi-line strings

```yaml
some-text: >
  Multiple
  lines
  of
  text
same-text: "Multiple lines of text\n"
```

YAML: Defining multi-line strings

```yaml
some-text: |
  Multiple
  lines
  of
  text
same-text: "Multiple\nlines\nof\ntext\n"
```

Setting up GitHub Workflows

- Each workflow is configured by a yaml file placed in `.github/workflows`
- Can be set to trigger by a wide variety of events
- Can run your own commands or use actions written by you or third-parties

Basic Structure of Workflow File

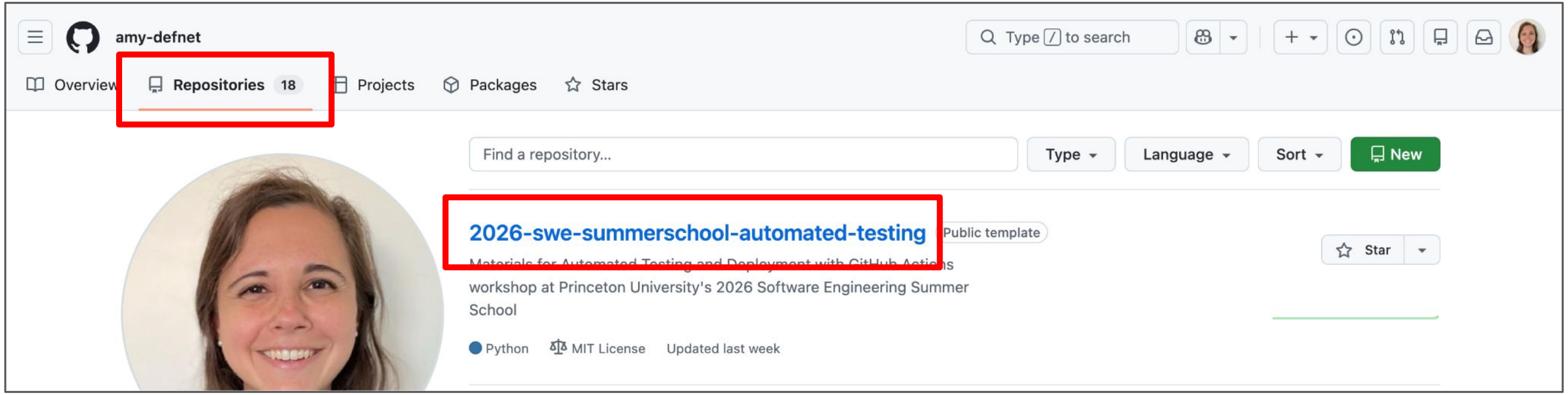
```yaml
on: <event or list of events>

jobs:
  job_1:
    name: <name of job>
    runs-on: <type of machine>
    steps:
      - uses: <some third-party action>
      - name: <name of step>
        run: <command>

  job_2:
    runs-on: <type of machine>
    steps:
      - uses: <command>
```

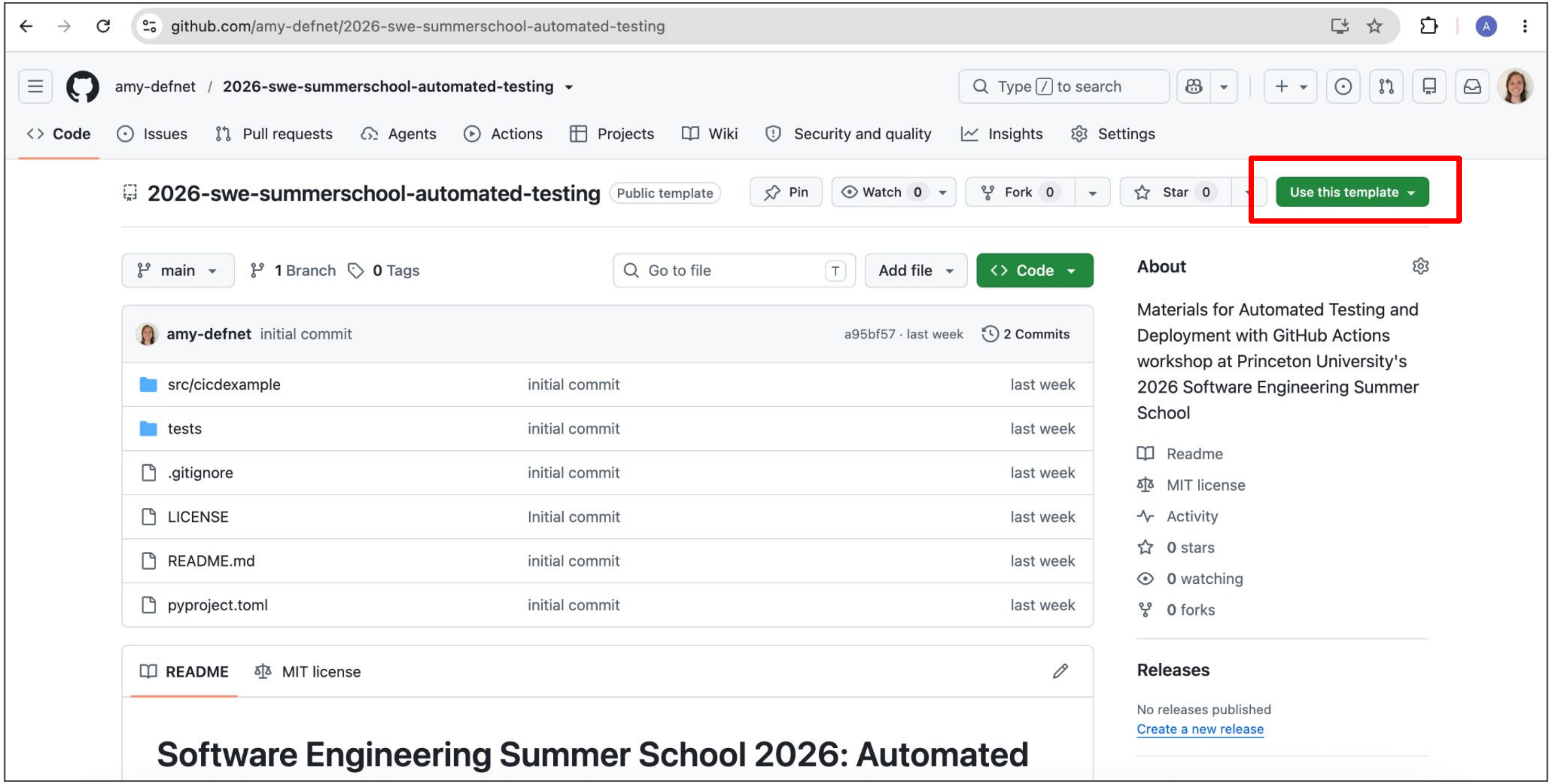
[GitHub-hosted runners for public repositories]()

Hands-On Exercises

Go to the following link:
https://github.com/amy-defnet/2026-swe-summerschool-automated-testing

Or https://github.com/amy-defnet → Repositories




github.com/amy-defnet/2026-swe-summerschool-automated-testing
amy-defnet / 2026-swe-summerschool-automated-testing
Type / to search
Code
Issues
Pull requests
Agents
Actions
Projects
Wiki
Security and quality
Insights
Settings
2026-swe-summerschool-automated-testing
Public template
Pin
Watch 0
Fork 0
Star 0
Use this template
main
1 Branch
0 Tags
Go to file
Add file
Code
amy-defnet initial commit
a95bf57 · last week
2 Commits
src/cicdexample initial commit last week
tests initial commit last week
.gitignore initial commit last week
LICENSE Initial commit last week
README.md initial commit last week
pyproject.toml initial commit last week
README
MIT license
Software Engineering Summer School 2026: Automated
About
Materials for Automated Testing and Deployment with GitHub Actions workshop at Princeton University's 2026 Software Engineering Summer School
Readme
MIT license
Activity
0 stars
0 watching
0 forks
Releases
No releases published
Create a new release

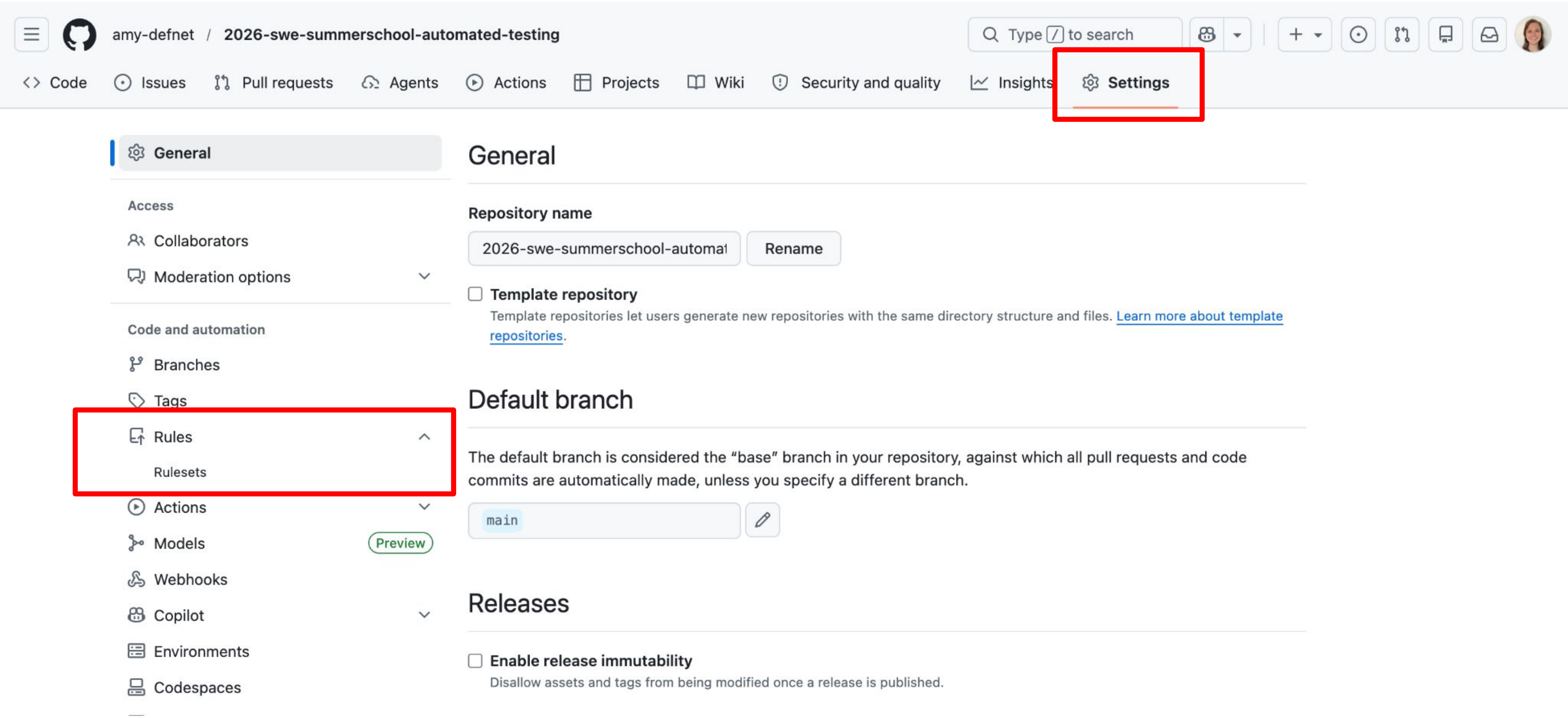
Exercise: Hello CI World

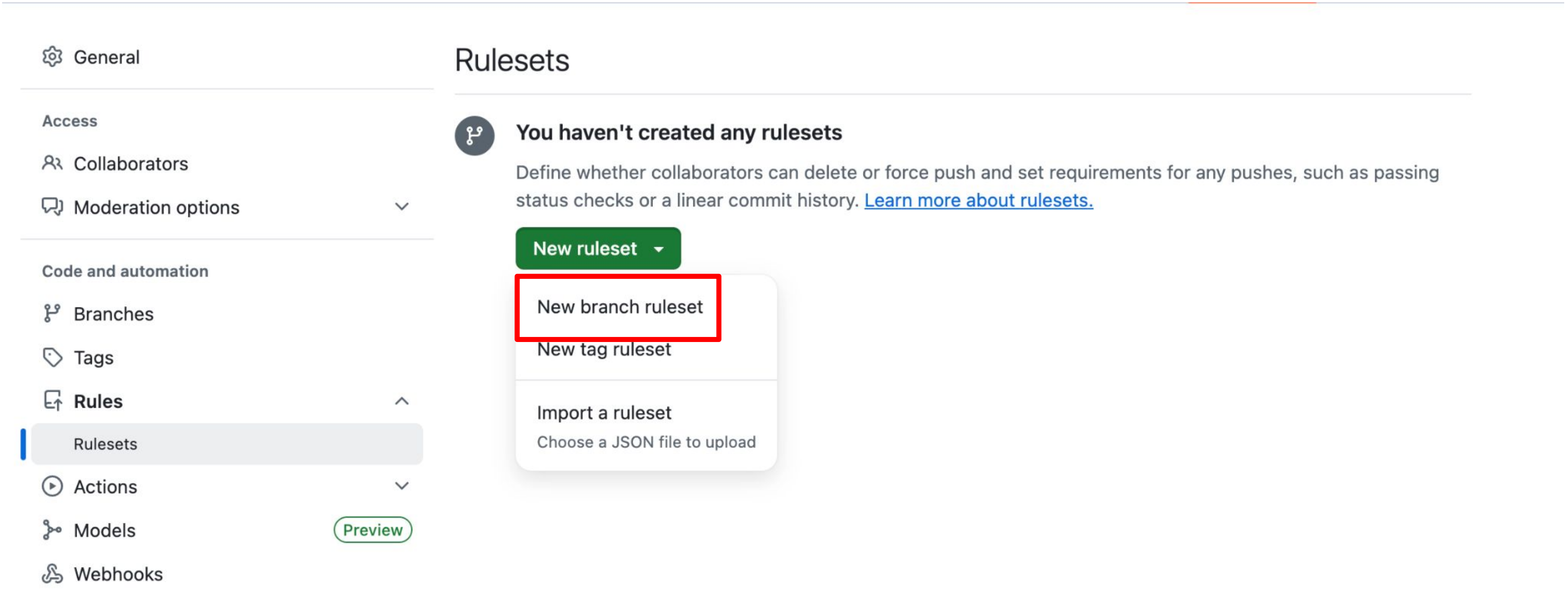
```yaml
name: Example
on: push
jobs:
  greeting:
    runs-on: ubuntu-latest
    steps:
      - name: Greeting!
        run: echo hello world
```

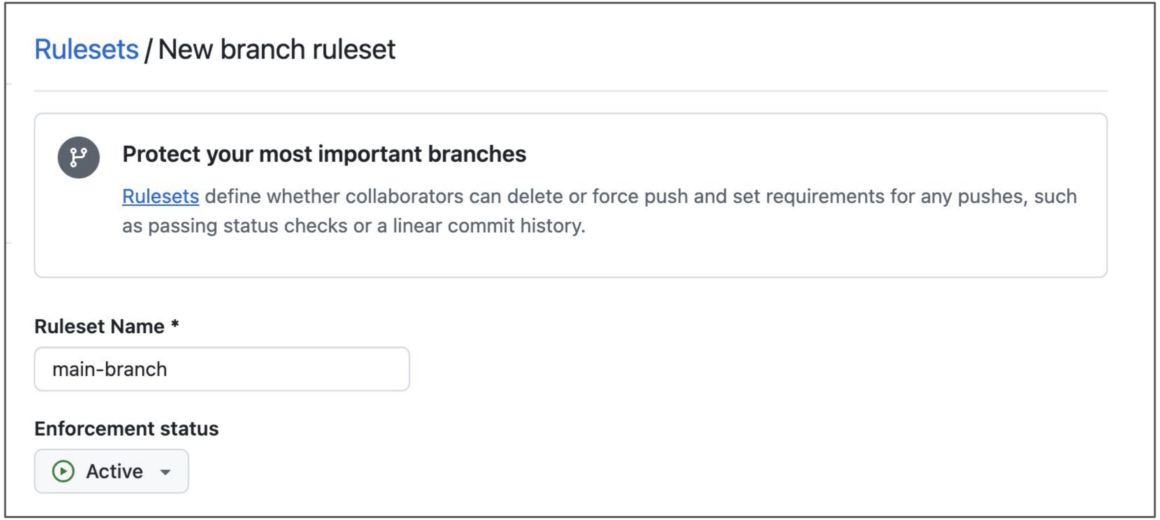
Exercise: CI for Python Testing

```yaml
name: Code Checks
on: push
jobs:
  greeting:
    runs-on: ubuntu-latest
    steps:
      - name: Greeting!
        run: echo hello world

  test-python-3-10:
    name: Check Python 3.10 on Ubuntu
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v3

      - uses: actions/setup-python@v4
        with:
          python-version: "3.10"

      - name: Install package
        run: python -m pip install -e .[test]

      - name: Test package
        run: python -m pytest
```

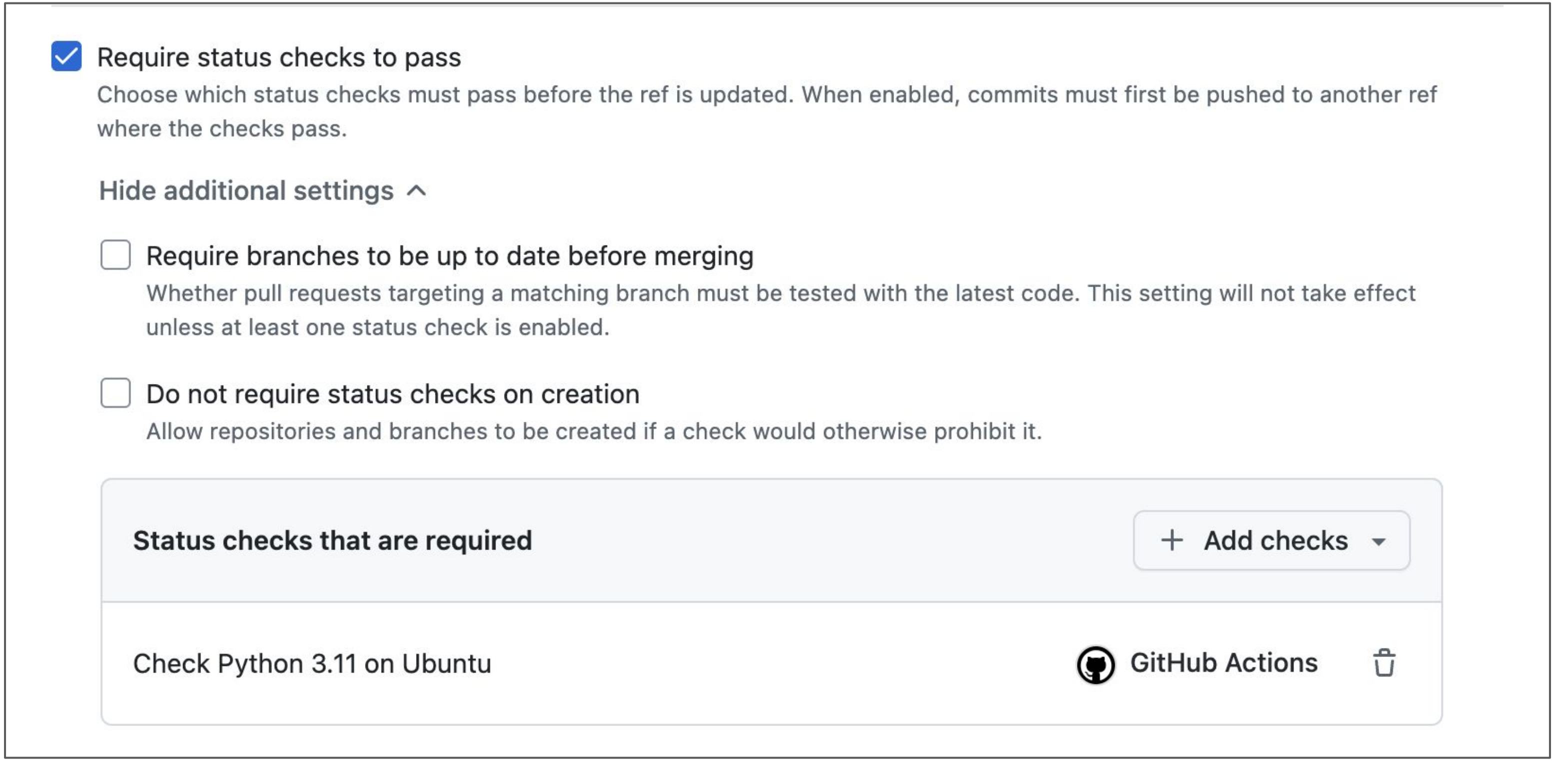
Exercise: Adding a Ruleset

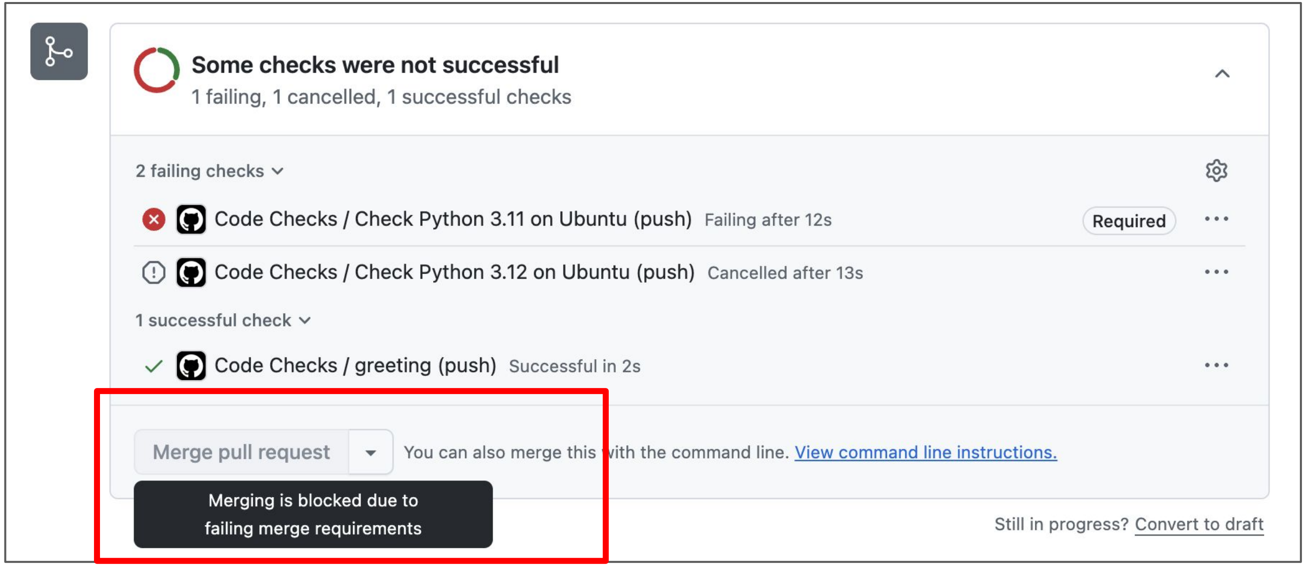

amy-defnet / 2026-swe-summerschool-automated-testing
Type / to search
Code
Issues
Pull requests
Agents
Actions
Projects
Wiki
Security and quality
Insights
Settings
General
Access
Collaborators
Moderation options
Code and automation
Branches
Tags
Rules
Rulesets
Actions
Models
Preview
Webhooks
Copilot
Environments
Codespaces
General
Repository name
2026-swe-summerschool-automat
Rename
Template repository
Template repositories let users generate new repositories with the same directory structure and files. Learn more about template repositories.
Default branch
The default branch is considered the "base" branch in your repository, against which all pull requests and code commits are automatically made, unless you specify a different branch.
main
Releases
Enable release immutability
Disallow assets and tags from being modified once a release is published.

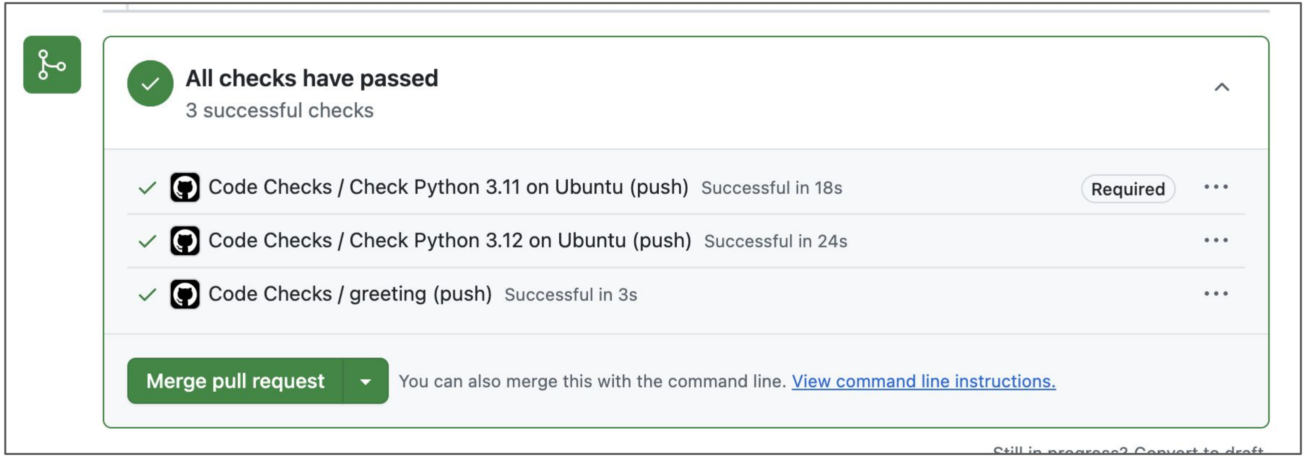

General
Access
Collaborators
Moderation options
Code and automation
Branches
Tags
Rules
Rulesets
Actions
Models
Preview
Webhooks
Rulesets
You haven't created any rulesets
Define whether collaborators can delete or force push and set requirements for any pushes, such as passing status checks or a linear commit history. Learn more about rulesets.
New ruleset
New branch ruleset
New tag ruleset
Import a ruleset
Choose a JSON file to upload


Rulesets / New branch ruleset
Protect your most important branches
Rulesets define whether collaborators can delete or force push and set requirements for any pushes, such as passing status checks or a linear commit history.
Ruleset Name *
main-branch
Enforcement status
Active


Target branches
Which branches should be matched?
Branch targeting criteria
Add target
main


Require status checks to pass
Choose which status checks must pass before the ref is updated. When enabled, commits must first be pushed to another ref where the checks pass.
Hide additional settings
Require branches to be up to date before merging
Whether pull requests targeting a matching branch must be tested with the latest code. This setting will not take effect unless at least one status check is enabled.
Do not require status checks on creation
Allow repositories and branches to be created if a check would otherwise prohibit it.
Status checks that are required
Add checks
Check Python 3.11 on Ubuntu
GitHub Actions


Some checks were not successful
1 failing, 1 cancelled, 1 successful checks
2 failing checks
Code Checks / Check Python 3.11 on Ubuntu (push) Failing after 12s
Required
Code Checks / Check Python 3.12 on Ubuntu (push) Cancelled after 13s
1 successful check
Code Checks / greeting (push) Successful in 2s
Merge pull request
You can also merge this with the command line. View command line instructions.
Merging is blocked due to
failing merge requirements
Still in progress? Convert to draft


All checks have passed
3 successful checks
Code Checks / Check Python 3.11 on Ubuntu (push) Successful in 18s
Required
Code Checks / Check Python 3.12 on Ubuntu (push) Successful in 24s
Code Checks / greeting (push) Successful in 3s
Merge pull request
You can also merge this with the command line. View command line instructions.

Exercise: Python Matrix

```yaml
name: Code Checks
on: push
jobs:

  test-python-versions:
    name: Check Python ${{ matrix.python-version }} on Ubuntu
    runs-on: ubuntu-latest
    strategy:
      matrix:
        python-version: ["3.10", "3.11"]
    steps:
      - uses: actions/checkout@v3

      - name: Setup Python ${{ matrix.python-version }}
        uses: actions/setup-python@v4
        with:
          python-version: ${{ matrix.python-version }}

      - name: Install package
        run: python -m pip install -e .[test]

      - name: Test package
        run: python -m pytest
```

Helpful Resources

- [GitHub Actions Documentation]

- [Weekly Research Computing Help Sessions]
  - Mondays 2:00-4:00pm
  - Wednesdays, Thursdays 1:00-3:00pm
  - Fridays 10:00am-12:00pm